

Assignment 5 – Monte Carlo for American Options

James Ezeilo

1) Introduction

American options differ from European options in that they can be exercised at any point up to and including maturity, which makes closed-form pricing unavailable for the most part and requires numerical methodology instead. Multiple approaches do exist to value the effect of this early-exercise feature, each with differing assumptions and sources of possible numerical error, so having an understanding of how they compare in practice is important to anyone working with real option data. In this study, I apply three pricing methods to SPY (S&P 500 ETF) American put options and evaluate how closely each one fits observed market mid-prices.

The purpose of this study is to compare the pricing accuracy of three models: the Black-Scholes European put (used as a benchmark), the Cox-Ross-Rubenstein (CRR) binomial tree, and the Longstaff-Schwartz Least-Squares Monte Carlo (LSM) method. All three are applied to the same real option chain retrieved via yfinance, with model accuracy being evaluated through the Root Mean Squared Error (RMSE) relative to observed market mid-prices. In addition to this, I examine how sensitive each method's accuracy is to its key numerical parameter – the number of steps for CRR and number of simulated paths for LSM.

The results show that the Black-Scholes produces the lowest RMSE despite being a European pricing model, while CRR and LSM are close to each other but slightly less

accurate in this specific sample. CRR accuracy is relatively stable across all step counts tested whereas LSM accuracy improves substantially as the number of paths increases from low levels, though the sampling noise will limit this improvement.

2) Data

The data used in this study were obtained in real time using the 'yfinance' library in Python. I collected the full option chain for SPY (S&P 500 ETF), focusing specifically on American put options.

The current underlying stock price S_0 was retrieved using the most recent daily closing price. The risk-free rate was obtained through the " IRX " ticker in yfinance and converted into a continuously compounded annual rate. SPY's dividend yield was also obtained directly through this library and used as a continuous dividend yield, q .

Out of the available expiration dates, I chose the first maturity to be at least 21 days from the current date. The time to maturity was computed as:

$$T = \frac{\text{Days to Expiration}}{365}$$

For each strike in the chosen expiration:

- Implied Volatility (IV) was extracted directly from the option chain.
- The Mid-Price (MP) was computed as:

$$\text{Mid Price} = \frac{\text{Bid} + \text{Ask}}{2}$$

These mid-prices are treated as market prices when evaluating model accuracy.

The final dataset, therefore, consists of strike price, time to maturity, implied volatility, market mid-price, risk-free rate, and dividend yield.

3) Methodology

This study compares 3 different pricing models:

- 1) **Black-Scholes** (European assumption)
- 2) **Cox-Ross-Rubinstein** (CRR) Binomial Tree
- 3) **Longstaff-Schwartz Monte Carlo** (LSM)

The accuracy of each model is evaluated using the Root Mean Squared Error (RMSE).

3.1 Implied Volatility Plot

Based on the chosen expiration, the implied volatility is plotted against strike price in order to visually display the volatility smile.

3.2 Black-Scholes European Put Pricing

Although these options are American, they are priced under the European assumption using the Black-Scholes model with a continuous dividend yield as a benchmark. The European put price is:

$$P = Ke^{-rt}N(-d_2) - S_0e^{-qT}N(-d_1)$$

Where:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

- S_0 - Current underlying price

- K - Strike price
- r - Risk-free rate
- q - Dividend yield
- σ - Implied volatility
- $N(\bullet)$ - Standard normal CDF

3.3 CRR Binomial Tree – American Put

The Cox-Ross-Rubenstein (CRR) binomial tree is used in this to price the American options allowing for early exercise. It contains 200 steps ($N = 200$).

Time step:

$$\Delta t = \frac{T}{N}$$

Up and down factors:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}$$

Risk-Neutral probability:

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}$$

At maturity, the payoff is:

$$V_T = \max(K - S_T, 0)$$

The tree is then rolled backwards using:

$$V = \max(e^{-r\Delta t}[pV_u + (1 - p)V_d], K - S)$$

The maximum operator ensures optimal early exercise at each node.

3.4 Longstaff-Schwartz Monte Carlo (LSM)

I also implement this American pricing by using the Longstaff-Schwartz least-squares Monte Carlo method.

First Step

The first step using this method is simulating risk-neutral paths under the risk-neutral measure:

$$S_{t+\Delta t} = S_t \exp \left(\left(r - q - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z \right)$$

Where $Z \sim N(0,1)$.

The baseline parameters used in this are:

- M = 20,000 simulated paths
- N = 50 exercise dates

Second Step

The next step is backward induction. At maturity:

$$CF_T = \max(K - S_T, 0)$$

For earlier exercise dates:

- 1) Identify in-the-money paths
- 2) Regress discounted continuation values on basis functions:

$$1, S_t, S_t^2$$

- 3) Estimate continuation value $\hat{C}_t$

- 4) Exercise if:

$$K - S_t > \hat{C}_t$$

The option price is equal to the discounted average optimal cash flow across simulations.

3.5 Model Accuracy – RMSE

Pricing Accuracy is evaluated using Root Mean Squared Error:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{i,Model} - P_{i,Market})^2}$$

RMSE is computed separately for Black-Scholes, CRR, and LSM. The method that exhibits the lowest RMSE is considered most accurate.

3.6 Sensitivity to Steps and Simulations

In order to analyse convergence, I vary both the CRR and the Monte Carlo (LSM) steps.

- CRR: N = 50, 100, 200, 500, 1000
- Monte Carlo: M = 100, 200, 500, 1000, 2000

For each specification, RMSE is recalculated to examine how pricing accuracy changes with model complexity.

4) Empirical Results

4.1 Implied Volatility (IV) Pattern

Figure 1 displays the volatility against the strike for the chosen SPY put expiration (2026-03-27). A clear strike dependence can be seen in implied volatility (the smile) rather than a flat volatility assumption. The IV appears to be higher for lower strikes and declines as strikes move upward through the central range, before flattening and slightly rebounding at the far-right tail. This pattern is consistent with the idea that market-implied tail risk is priced differently across moneyness, so a single constant volatility parameter would not describe the entire option surface well.

4.2 Model Prices vs Market Mid Prices

Table 1 displays the market mid prices along with the model-implied prices from the Black-Scholes European put, the CRR American put (binomial tree), and the LSM American put (Monte Carlo) from the first 20 strike prices. Across all the strikes shown, all 3 models generally move in line with the market mid prices, increasing as the strike increases (i.e., as the puts become more in-the-money). This indicates that each model is capturing the main relationship between strike and put value for this expiration.

The Black-Scholes price, in terms of relative fit, often trails below the market mid prices across the majority of the range, but the gap tends to be small and fairly consistent. The CRR and LSM are usually close to each other, which is expected since both methods are designed to price the American-style option, and they frequently produce similar valuations when early-exercise value is limited. Where differences appear, such as at

the strike price of 590 where the price is only \$0.01 off each other. This reflects discretisation effects in the tree and simulated/regression noise in LSM rather than a divergence in one direction.

Overall, the table suggests that the market pricing of this chain is consistent with standard no-arbitrage valuation frameworks. However, the small deviations between model and mid prices also highlight the fact that real-world option quotes include further features that aren't captured by my methods. An example of this could be from volatility surface effects that go beyond using the point-in-time implied volatility for each strike.

4.3 RMSE Comparison Across Pricing Methods

This section evaluates the pricing accuracy across the full option chain through the computation of the Root Mean Squared Error (RMSE) between each model's price and the market mid-price across all strikes (Table 2). A lower RMSE indicates that a model's prices are closer to observed market quotes on average.

Based on the Table 2 results, the Black-Scholes is the most accurate method for this specific chain since it produces the lowest RMSE (average pricing error) at 0.424516. Even though Black-Scholes assumes European exercise, it fits the mid prices better than the two American-pricing approaches in this sample. It can be interpreted as, for this given expiration and market environment, the incremental early-exercise value captured by CRR and LSM is either relatively small or not strongly reflected in mid-prices once bid-ask effects are considered. Furthermore, the CRR method can

introduce discretisation error from the time-step grid, while the LSM method adds simulation noise and regression approximation error when estimating continuation values. These implementation-based errors can reduce fit even though the models are theoretically more flexible for American options.

Overall, Table 2 shows that the more complex American-pricing methods do not necessarily outperform the European benchmark in terms of empirical fit. This highlights that model accuracy in practice doesn't only depend on theoretical option features (like early exercise), but also on how much numerical error the method introduces and how closely its assumptions match the way real option quotes are observed in the market.

4.4 Sensitivity to CRR Steps and Monte Carlo Simulations

To examine how numerical settings affect pricing accuracy, two sets of tests have been conducted. The first test is where the number of steps in the CRR binomial tree is varied, and the second test varies the number of simulated paths in the LSM Monte Carlo method, computing RMSE for each case. The values of the results from each of these tests can be seen in Table 3, with a graphical representation present in Figure 2.

For the CRR model, step counts of $N \in \{50, 100, 200, 500, 1000\}$. As shown in the left graph of Figure 2, the RMSE values stay very close together across all specifications, clustering around the 0.51 range (e.g., 0.5154 at $N = 50$ vs 0.5132 at $N = 1000$). This indicates that once the tree has a reasonable number of steps, adding more does not

meaningfully improve the fit, and the remaining pricing error is likely driven by something other than the step size itself.

For the LSM Monte Carlo method, I vary the number of paths over $M \in \{100, 200, 500, 1000, 2000\}$ while holding the number of exercise dates fixed at $N = 50$.

The right graph of Figure 2 visualises the sharp initial drop in RMSE – starting at 1.2305 for $M = 100$, falling to 0.6031 at $M = 200$, and further to 0.5001 at $M = 500$. The decline is not perfectly smooth since RMSE rises slightly to 0.5751 at $M = 1000$ before falling again to 0.4874 at $M = 2000$. This reflects the randomness that is inherent in Monte Carlo sampling as well as the regression approximation LSM uses for continuation values.

Overall, these results show that LSM accuracy improves significantly as the path count grows from low levels, while CRR accuracy is relatively stable once the tree has reached a reasonable number of steps.

5) Conclusion

This study compared Black-Scholes (benchmark), CRR binomial tree, and LSM Monte Carlo for pricing SPY American put options, evaluating accuracy against market mid-prices using RMSE. The implied volatility plot confirms a clear smile across strikes, and all three models closely track market prices across the strike range.

Although the Black-Scholes is priced under a European assumption, it produced the lowest RMSE of 0.4245, outperforming both CRR and LSM. This suggests that for this expiration and market environment, the theoretical advantage of modelling early-exercise does not translate into better empirical fit, as numerical errors from tree discretisation and Monte Carlo regression partially offset the gains from a more flexible framework.

The sensitivity analysis also shows that CRR accuracy is mostly unaffected when the number of steps increases once the tree reaches a reasonable number of them, while the LSM is more sensitive at low simulated path counts. This can be seen in the RMSE when it falls sharply from 1.2305 at $M = 100$ to around 0.49 at $M = 2000$.

Overall, the results highlight that model complexity does not guarantee better fit in practice, and that numerical implementation choices do matter when pricing performance is evaluated against real market data.

6) Appendix

Figure 1: Implied Volatility

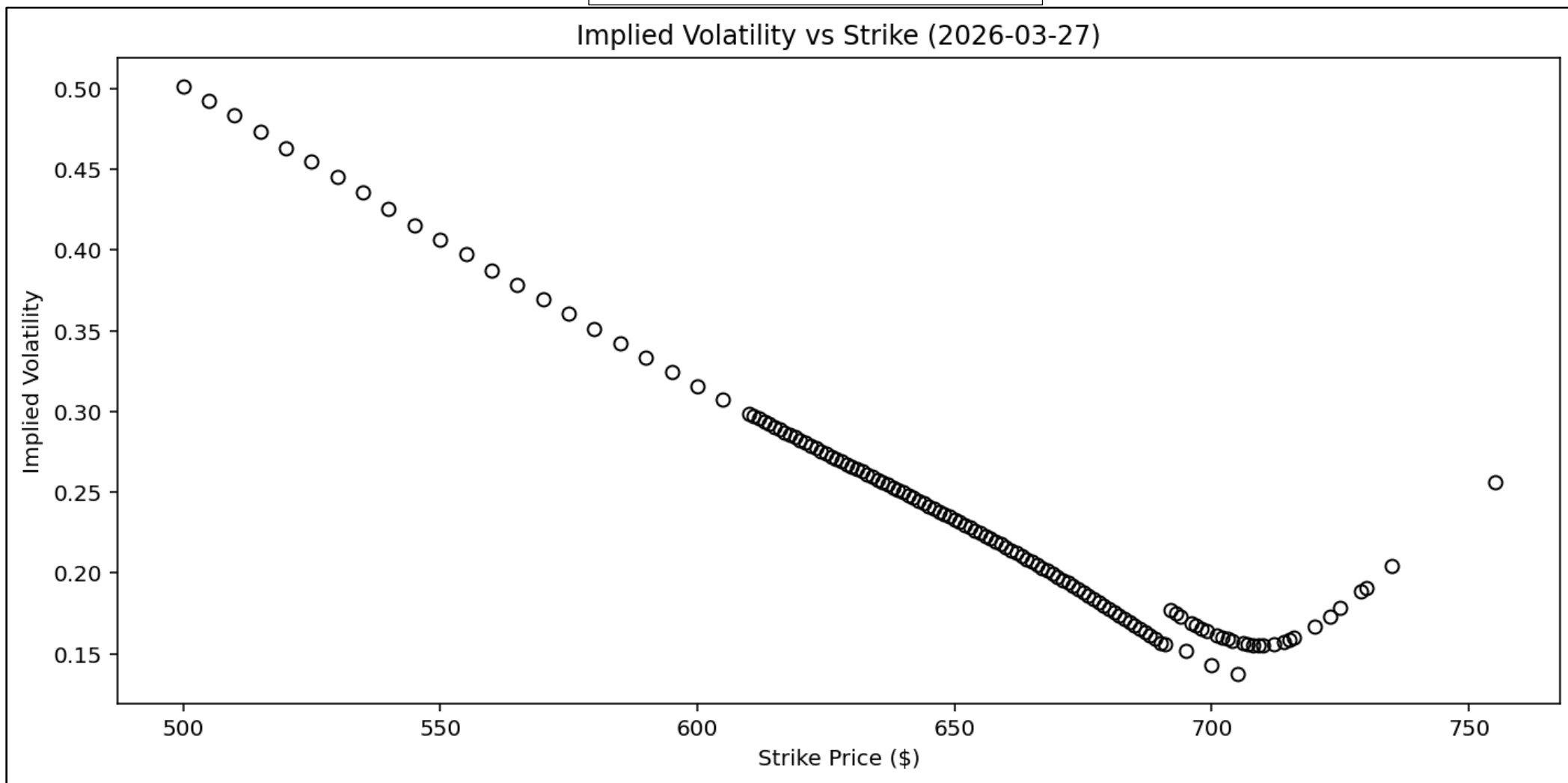


Figure 2: Sensitivity Analysis: RMSE vs Steps/Paths

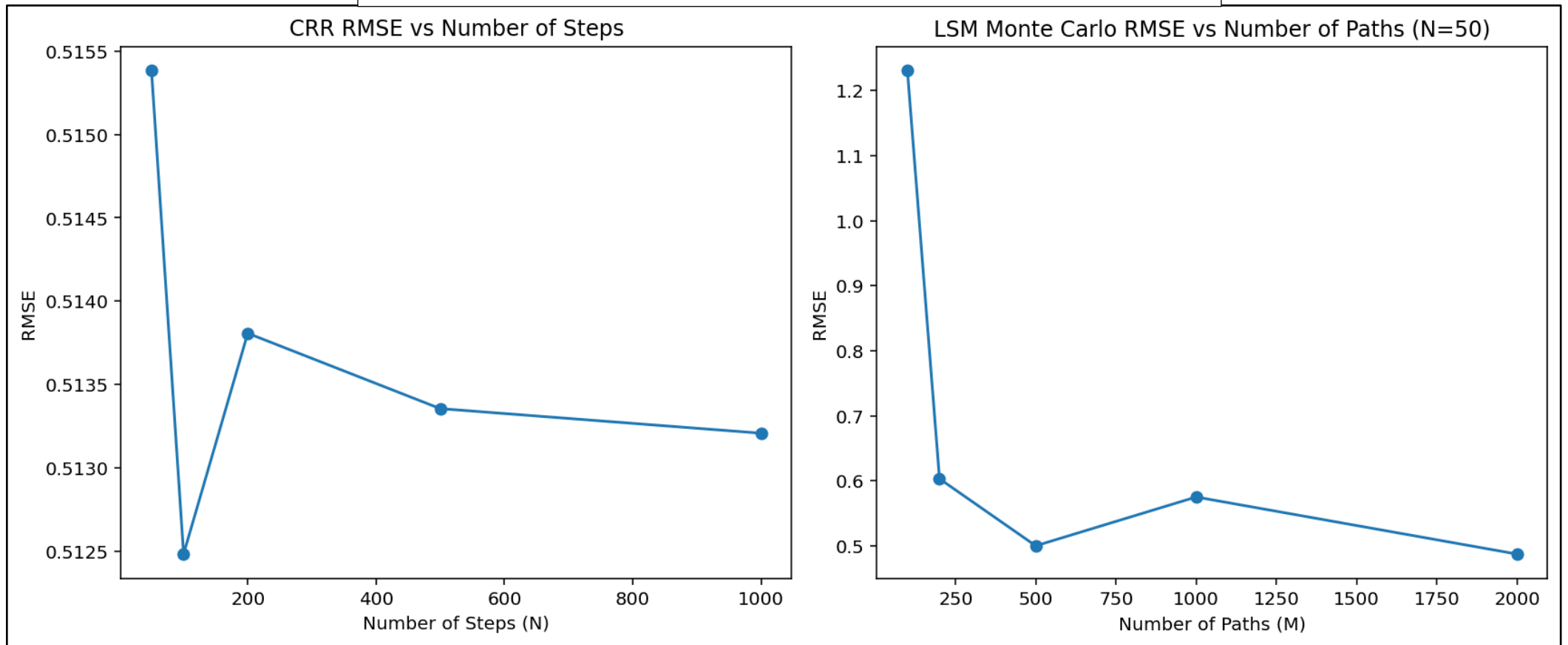


Table 1: Model Prices vs Market Mid-Prices

Strike (K)	Mid-Price	Black-Scholes	CRR Binomial Tree	LSM Monte Carlo
500	0.235	0.226314	0.223854	0.224671
505	0.255	0.248723	0.24695	0.241569
510	0.275	0.269079	0.265838	0.262927
515	0.295	0.286581	0.280471	0.280682
520	0.315	0.306476	0.302393	0.301948
525	0.345	0.335717	0.334151	0.327733
530	0.37	0.363538	0.361341	0.357349
535	0.4	0.392288	0.386641	0.379203
540	0.435	0.420206	0.414947	0.414312
545	0.465	0.448601	0.446966	0.436861
550	0.505	0.486924	0.485666	0.477569
555	0.55	0.534205	0.528352	0.518601
560	0.595	0.572897	0.567805	0.561974
565	0.65	0.628529	0.628357	0.615734
570	0.71	0.684949	0.682951	0.66435
575	0.78	0.751786	0.743067	0.728122
580	0.85	0.818325	0.818399	0.796312
585	0.94	0.904302	0.904054	0.885352
590	1.04	0.997091	0.987653	0.975041
595	1.15	1.101375	1.103746	1.0806

Table 2: RMSE Comparison

Method	RMSE (vs Mid-Price)	Most Accurate?
Black-Scholes (European)	0.424516	Yes (Lowest RMSE)
CRR Binomial Tree (American)	0.513807	No
LSM Monte Carlo (American)	0.521568	No

Table 3: Sensitivity Analysis

Model	N (Steps)	RMSE
CRR	50	0.515382
CRR	100	0.512483
CRR	200	0.513807
CRR	500	0.513355
CRR	1000	0.513208

Model	M (Paths)	N (Exercise Dates)	RMSE
LSM MC	100	50	1.230454
LSM MC	200	50	0.603122
LSM MC	500	50	0.500091
LSM MC	1000	50	0.575143
LSM MC	2000	50	0.487423